

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. Microscopic states, Koopman observables, push-forward laws, ensemble log-Laplace values, and the limiting kinetic semigroup are kept distinct. An observable resolvent core and full-hierarchy terminal correctors give the cylinder Hamiltonian, while the B2 dynamic action constructs the additive Lax–Oleinik semigroup without repeated preparation cost. A two-scale coercive Tataru comparison and the prepared Laplace principle identify the microscopic limit.

Platform identifier: HSBG-LI0-v3. **Controlling revision:** ROUND10-REFEREE-POSITIVE-CLOSURE.
Registered proof source: B4_DYNAMIC_ACTION_GRAPH_CORE_COMPARISON.tex.

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1. ROUND-TEN RECONSTRUCTION: TYPED MICROSCOPIC OBJECTS, A GRAPH CORE, AND THE KINETIC ACTION SEMIGROUP

Five objects are kept distinct: the microscopic state, the deterministic Koopman semigroup on microscopic observables, its dual push-forward on laws, finite-horizon log-Laplace values under an initial ensemble, and the limiting nonlinear semigroup on kinetic densities. The limiting semigroup is obtained from the proved B2 action and the Laplace principle, not by applying a state resolvent to an observable.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 49L25, 60F10; Secondary 82C40, 35D40.

Key words and phrases. kinetic Hamilton-Jacobi equation, nonlinear semigroup, micro-canonical preparation, theta-expectation.

1.1. Microscopic state, observable, and law semigroups. Let E_ε be the graph-completed hard-sphere phase space augmented by the cumulative actual-contact ledger, and let $\Phi_{s,t}^\varepsilon$ be the deterministic flow. Define

$$P_{s,t}^\varepsilon F(x) = F(\Phi_{s,t}^\varepsilon x), \quad P_{s,t}^{\varepsilon*} \rho = (\Phi_{s,t}^\varepsilon)_\# \rho.$$

The first acts on observables, the second on laws. If G is a terminal macro-observable and ρ an initial ensemble, put

$$\mathcal{V}_{s,t}^\varepsilon(G; \rho) = \mu_\varepsilon^{-1} \log \int_{E_\varepsilon} e^{\mu_\varepsilon G(X_t^\varepsilon(x))} d\rho(x).$$

Proposition 1.1 (Typed finite-horizon tower). *The Koopman and push-forward families satisfy their exact dual towers. The finite-horizon value obeys the Bayes tower*

$$\mathcal{V}_{r,t}^\varepsilon(G; \rho) = \mathcal{V}_{r,s}^\varepsilon \left(x \mapsto \mathcal{V}_{s,t}^\varepsilon(G; \delta_x); \rho \right),$$

with the inner function understood before the outer logarithm. No law functional is evaluated at a microscopic point, and no push-forward resolvent is applied to an observable.

Proof. Both linear towers follow from $\Phi_{r,t} = \Phi_{s,t} \circ \Phi_{r,s}$. Disintegrate the initial integral at time s and apply the elementary identity $e^{\mu \mathcal{V}_{s,t}^\varepsilon(G; \delta_x)} = P_{s,t}^\varepsilon(e^{\mu G})(x)$. This proves the displayed formula with all domains typed. $\square$

1.2. A genuine observable graph core. Let A_ε be the closed generator of the Koopman semigroup on the weighted observable space. For $\lambda > 0$ define the observable resolvent

$$R_{\lambda,\varepsilon}^{\text{obs}} F = \int_0^\infty e^{-\lambda t} P_{0,t}^\varepsilon F dt.$$

Let $\mathcal{D}_0$ be the smooth weighted configuration/contact cylinders satisfying the specular matching condition, and set

$$\mathcal{C}_\varepsilon = \text{span}\{R_{\lambda,\varepsilon}^{\text{obs}} F : F \in \mathcal{D}_0, \lambda \in \mathbb{Q}_+\}.$$

Theorem 1.2 (Observable resolvent core). $\mathcal{C}_\varepsilon \subset D(A_\varepsilon)$ and

$$(\lambda - A_\varepsilon) R_{\lambda,\varepsilon}^{\text{obs}} F = F.$$

Its closure in the graph norm is $D(A_\varepsilon)$. Smooth bounded functional calculus on finite collections of core functions preserves the specular graph domain. No entire cutoff is used.

Proof. The B2 reflected transport semigroup is strongly continuous on observables, so Hille–Yosida gives the resolvent identity. The range of a fixed $\lambda - A_\varepsilon$ on the dense cylinder class is dense in the observable space; applying the bounded inverse proves graph-core density. The ordinary chain rule holds on smooth compatible cylinders, while specular matching is preserved because both one-sided traces have equal cylinder values and equal tangential derivatives. Approximation in the resolvent range passes this property to the graph closure. $\square$

1.3. Finite-generator correctors and the cylinder Hamiltonian. For a kinetic cylinder F let $F \circ X^\varepsilon$ be its microscopic lift. Expand the B2 marked pressure on a deterministic time block and denote by $D_{j,\varepsilon}$ the connected j -label defect after lower orders have been removed. The full finite hierarchy is triangular on the factorial label scale; let $V_\varepsilon(t, s)$ be its backward observable evolution, not a closed order- j evolution.

Lemma 1.3 (Full-hierarchy terminal corrector). *The vector corrector*

$$c_\varepsilon(t) = \int_t^T V_\varepsilon(t, s) D_\varepsilon(s) ds$$

belongs to the product observable graph domain, has zero terminal value, and solves

$$\partial_t c_\varepsilon + A_\varepsilon(t) c_\varepsilon = -D_\varepsilon(t).$$

Its factorial connected expansion and the expansion of its generator image are normally summable on a slightly smaller label weight.

Proof. B2's positive boundary-renewal hierarchy constructs the full product evolution by Picard iteration. The factorial estimate $j!C^jT^{j-1}$ is summable after multiplication by μ_ε^{1-j} and one loss of label weight. Differentiating the Duhamel integral gives the equation in the closed graph domain. The equation itself expresses the generator image as a normally summable defect plus a time derivative, so the infinite corrector may be interchanged with the finite generator. $\square$

Theorem 1.4 (Cylinder Hamiltonian convergence). *For every bounded-increment kinetic cylinder there are corrected microscopic observables $F_\varepsilon \in D(A_\varepsilon)$ such that the finite-volume source derivative of the logarithmic ensemble value converges locally uniformly to*

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \int (e^{\Delta p + \psi} - 1) dA_f.$$

The statement concerns ensemble log-Laplace derivatives and does not identify the deterministic push-forward on laws with a nonlinear Markov generator.

Proof. Insert the full-hierarchy corrector into the exact B2 connected source expansion. It cancels every correlation defect in the deterministic block. Creation trees converge by collision coarea to the displayed exponential, while the integrated first-surplus estimate removes cycles. Normal complex convergence permits the source derivative and corrector sum to pass to the limit. $\square$

1.4. The dynamic action semigroup. For a balanced pair on $[s, t]$ define only the dynamic cost

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f.$$

The prepared initial rate is charged once at time zero.

Theorem 1.5 (Additive kinetic Lax–Oleinik semigroup). *For bounded uniformly continuous terminal data,*

$$(S_{s,t}\Phi)(f_s) = \sup_{(f,\Gamma):f(s)=f_s} \{\Phi(f_t) - \mathcal{A}_{s,t}(f, \Gamma)\}$$

is finite on moment sublevels, has compact relaxed maximizer sets, and satisfies $S_{r,t} = S_{r,s}S_{s,t}$. The prepared value is

$$\sup_{f_0} \{S_{0,T}\Phi(f_0) - I_{0,a}(f_0)\}.$$

Proof. B2 gives compact action sublevels and closed balance. Restriction and concatenation of a balanced pair preserve the intermediate trace and add the two entropy integrals, proving both dynamic-programming inequalities. The initial rate is absent from every transition interval and appears only in the last displayed boundary optimization. $\square$

1.5. Two-scale comparison. For $M > 0$, restrict $q = d\Gamma/dA_f$ to $e^{-M} \leq q \leq e^M$ and denote the Hamiltonian by $\mathbb{H}_M$. Let d_λ be the Tataru distance generated by the resolvent core truncated at slope λ .

Theorem 1.6 (Coercive diagonal comparison). *For fixed M , bounded upper and lower viscosity solutions of*

$$\partial_t U + \mathbb{H}_M(f, DU) = 0$$

satisfy comparison. One may use the penalty

$$\frac{d_\lambda(f, g)^2}{2\epsilon} + \delta(\Upsilon(f) + \Upsilon(g)),$$

first sending $\epsilon \downarrow 0$ at fixed λ , and only then $\lambda \uparrow \infty$. The first limit forces the diagonal; the second recovers the full source topology without ever evaluating the Hamiltonian on an uncontrolled test jet.

Proof. At a doubled maximum the coercive quadratic penalty implies $d_\lambda(f, g) = O(\sqrt{\epsilon})$. Its derivative belongs to the finite resolvent core and has slope bounded by λ . For fixed (M, λ) , Hamiltonian continuity on a common moment sublevel is uniform, so subtraction of the viscosity inequalities and $\epsilon \downarrow 0$ gives the contradiction. Monotonicity of d_λ and density of the resolvent core then permit $\lambda \uparrow \infty$; the Lyapunov term is removed last. No post hoc saturation changes the maximizing penalty. $\square$

Theorem 1.7 (Untruncated semigroup and microscopic Laplace limit). *The truncated semigroups S^M converge locally uniformly to S as $M \rightarrow \infty$, comparison passes to the entropy-coercive full Hamiltonian, and for every prepared regular initial family*

$$\mathcal{V}_{0,T}^\epsilon(\Phi \circ X^\epsilon; \rho_\epsilon) \longrightarrow \sup_{f_0} \{S_{0,T}\Phi(f_0) - I_{0,a}(f_0)\}.$$

Proof. Entropy coercivity makes the action carried by $q \notin [e^{-M}, e^M]$ vanish uniformly on compact action sublevels after truncation and exact B2 balance repair. Hence $S^M \rightarrow S$ locally uniformly; stability of viscosity solutions and Theorem 1.6 give full comparison. The final convergence is the Laplace principle for the B1–B2 prepared good LDP. It uses the correctly typed finite-horizon ensemble value from Proposition 1.1, not an ill-typed law semigroup. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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